



**PRACTICE Midterm Exam
Fall 2025**

YOUR NAME: _____

YOUR SEMINAR INSTRUCTOR'S NAME: _____

--- This is a double-sided exam ---

Welcome to the Midterm Exam for Frontiers of Science. Please **write your name and your seminar instructor's name on this exam**. The exam begins at **2 pm** and finishes at **4 pm**, giving you **2 hours** to complete it.

This exam consists of four questions: two Mind and Brain questions (15 points each; parts A, B & C) and two Physics and Reality questions (15 points each; parts A, B & C). You must answer ALL questions in the exam packet within the empty space that follows each question. Should you need extra space, you may use the empty pages in the back of the exam. Feel free to use your calculator as necessary. To receive full credit, show all your work and include units where applicable. Be sure to turn in this exam packet!

As a guideline for your answers, each question is marked with: [NC] – 'requires no calculation', [SC] – 'requires simple calculations or quantitatively reading a graph' or [C] – 'requires calculations'.

Numerical Constants, Values, and Unit Conversions

c (Speed of light)	$3.0 \times 10^8 \text{ m s}^{-1}$
h (Plank's constant)	$6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$
G (Gravitational constant)	$6.7 \times 10^{-11} \text{ m}^3 \text{ s}^{-2} \text{ kg}^{-1}$
R_E (Radius of Earth)	$6.4 \times 10^6 \text{ m}$
M_E (Mass of Earth)	$6.0 \times 10^{24} \text{ kg}$
M_S (Mass of Sun)	$2.0 \times 10^{30} \text{ kg}$
r_p (Radius of proton)	$8.4 \times 10^{-16} \text{ m}$
r_H (Radius of hydrogen atom)	$5.3 \times 10^{-11} \text{ m}$
m_p (Mass of proton)	$1.7 \times 10^{-27} \text{ kg}$
m_e (Mass of electron)	$9.1 \times 10^{-31} \text{ kg}$

Metric System Prefixes

T	"tera"	10^{12}
G	"giga"	10^9
M	"mega"	10^6
k	"kilo"	10^3
c	"centi"	10^{-2}
m	"milli"	10^{-3}
μ	"micro"	10^{-6}
n	"nano"	10^{-9}
p	"pico"	10^{-12}

$$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 6.24 \times 10^{18} \text{ eV}$$

$$1 \text{ mile} = 1.61 \text{ km}$$

$$1 \text{ foot} = 0.305 \text{ m}$$

$$1 \text{ parsec (pc)} = 3.26 \text{ light years (ly)} = 3.09 \times 10^{16} \text{ m}$$

$$1 \text{ metric ton} = 1000 \text{ kg}$$

$$1 \text{ lb} = 0.454 \text{ kg}$$

$$1 \text{ L} = 10^{-3} \text{ m}^3$$

$$1 \text{ gallon} = 3.79 \text{ L}$$

$$\pi = 3.14$$

Distance

$$d = v t$$

d = distance, v = velocity, t = time

Gamma factor

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

γ = gamma factor, v = velocity (speed), c = speed of light

Time dilation and length contraction

$$(\text{time elapsed for moving object}) = \frac{(\text{time elapsed for stationary object/observer})}{\gamma}$$

$$(\text{length of moving object}) = \frac{(\text{length of stationary object})}{\gamma}$$

Energy of a massive particle

$$E = \gamma m_r c^2$$

m_r = rest mass of object, c = speed of light

Schwarzschild radius

$$r_s = \frac{2GM}{c^2}$$

r_s = Schwarzschild radius, G = gravitational constant, M = mass of object, c = speed of light

Gravitational time dilation

$$T = t \times \left(1 - \frac{2GM}{rc^2}\right)^{1/2}$$

T = time at a distance r from a massive object's center, t = time in empty space, G = gravitational constant, M = mass of object, c = speed of light, r = distance from a massive object's center

De Broglie Wavelength

$$\lambda = \frac{h}{mv}$$

λ = wavelength, h = Planck's constant, m = mass of object, v = speed

Heisenberg Uncertainty Principle

$$\Delta x \Delta v \geq \frac{h}{4\pi m}$$

Δx = uncertainty in position, Δv = uncertainty in velocity (speed), h = Planck's constant, m = mass of object

Geometry

$$V_{\text{sphere}} = \frac{4}{3} \pi r^3$$

$$V_{\text{cylinder}} = h \pi r^2$$

V = volume, r = radius

V = volume, r = radius, h = height

Density

$$D = \frac{M}{V}$$

D = density, M = mass of object, V = volume

Statistics

$$SE = \frac{SD}{\sqrt{N}}$$

SE = standard error of the mean, SD = standard deviation,

N = sample size

MIND AND BRAIN

Question 1. [15 points]

In a 2016 study, Setlow and co-workers performed a series of experiments using rats to test the hypothesis that there are sex differences in decision making involving risk of punishment. The scientists also wanted to examine whether amphetamine intake affects the tendency to take risks.

1A. [5 points, SC]

In one set of experiments, a group of female rats and a group of male rats had the option to press one of two levers. One of the levers presented a small food reward, while the other presented a larger food reward, but with a preset probability that the rat would receive a punishment in the form of a brief foot shock prior to the reward. The experiments were conducted in several blocks of trials in which different punishment probabilities were tested (0%, 25%, 50%, 75%, and 100%). At the beginning of each block of trials there was a learning period to teach the rats about the rewards associated with the two levers and the punishment probability associated with the larger reward. The results of this first set of experiments are illustrated in Figure 1A below.

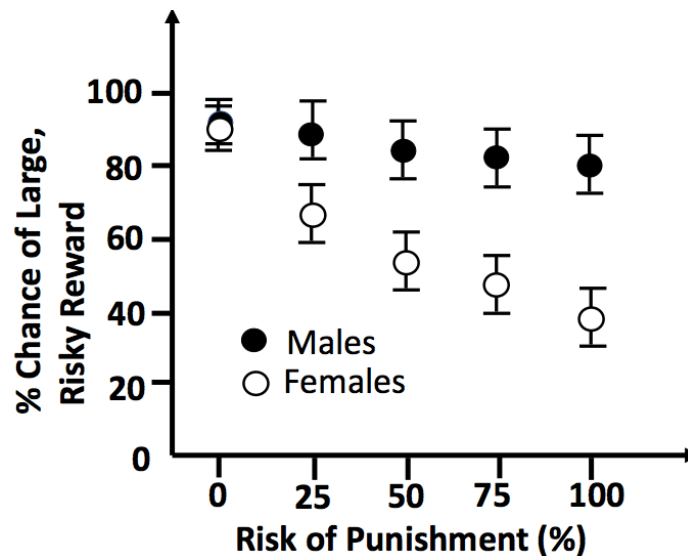


Figure 1A. Performance in the risky decision making task of male and female rats. The y-axis represents the percentage of instances that rats chose the large reward for each block of trials, averaged over all male or all female rats. Error bars represent ± 1 SE.

1A. (i) According to Figure 1A, were there sex differences in decision making, and if so, under what conditions? Explain using your knowledge of statistics.

1A. (ii) What do these findings suggest about the hypothesis being tested by the scientists? Explain in 1-2 sentences.

1A. (iii) According to Figure 1A, what is the relationship between percent choice of large risky reward and risk of punishment in male rats and in female rats? Explain using your knowledge of statistics.

1B. [5 points, C]

Amphetamine is a stimulant known to target dopamine neurons. In a second set of experiments, the authors administered different doses of amphetamine to the female and male rats and performed the same tests described above. The results of these experiments for the male rats are reported in Figure 1B below.

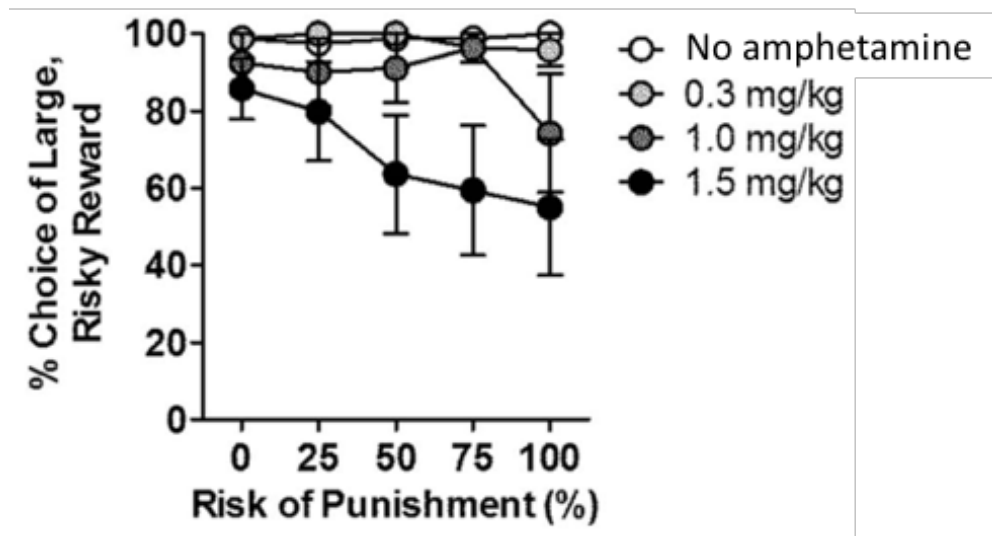


Figure 1B. Effect of amphetamine on performance in the risky decision making task for male rats. Error bars represent 95% confidence intervals.

1B. (i) Why might scientists expect amphetamine to have an effect on the rat's decision making?

1B. (ii) Suppose that you give a group of male rats one eighth of a 5 mg capsule of Adderall medication. About 62% of the weight of an Adderall capsule is amphetamine. The average weight of a rat is ~260 g. **For such rats, estimate the 95% confidence interval of % Choice of Large, Risky Reward when the risk of punishment is 50%.**

1C. [5 points, NC]

These rat experiments are an example of decision making informed by reward prediction errors.

1C. (i) In 2-3 sentences describe one experiment that you learned about in lecture, which was used by scientists to learn about this type of decision making.

1C. (ii) What part of the brain is involved in this type of decision making in humans?

1C. (iii) Imagine that another research lab decides to study whether there are any sex differences in decision making in humans, but half of the female participants are affected by Parkinson's disease. **What type of error would this introduce (random or systematic), and how would this error affect the accuracy and precision of this experiment?** Explain in 2-3 sentences.

Question 2. [15 points]

There has been recent interest in the hypothesis that chewing gum enhances memory. Dr. Brittney Floyd and her colleagues designed an experiment to test this hypothesis. They set out to determine whether the effects of gum chewing on memory are related to attention and how long any such effects last after chewing.

Participants in the experiment chewed gum for five minutes, then encoded a list of words, and later were tested on their recall of the words. The encoding phase was conducted under two different conditions: Full Attention or Divided Attention. In full attention encoding, participants were told to try and remember a list of words. In divided attention encoding, participants were told to try and remember a list of words but also to simultaneously press random computer keys. Figure 1A (below) shows the effects of chewing gum on memory retrieval for full and divided attention encoding. Early and Late conditions refer to the timing of the task (encoding and retrieval) relative to the gum chewing session: the early condition refers to the task taking place just after chewing and discarding the gum, whereas the late condition refers to the task taking place approximately 20 minutes later.

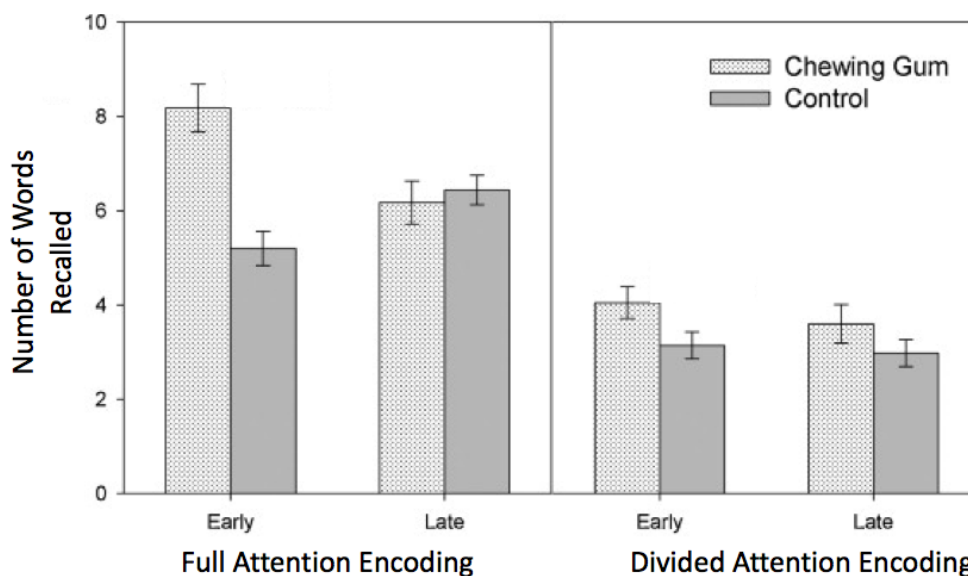


Figure 2A. Mean number of words recalled by participants who performed the task early and late (relative to gum chewing), and who chewed gum (chewing gum, light gray bars) or did not chew gum (control, dark gray bars) prior to the experiment. Left panel shows data from the full attention encoding experiments, and right panel shows data from the divided attention encoding experiments. Error bars represent 95% confidence intervals.

2A. [5 points, SC]

2A. (i) According to the data presented in Figure 2A above, did chewing gum affect recall in each of the four experiments (full attention and divided attention encoding; early and late testing)? Justify your answers using your knowledge of statistics. Provide your answer on the next page.

2A. (ii) Do the experimental results in Figure 2A support the hypothesis that chewing gum improves memory? Explain in 1-2 sentences.

2A. (iii) Several recent observational studies have found a statistically significant correlation between the number of missing teeth and hippocampus-dependent cognitive function in older adults. Some researchers have suggested that a decreased chewing ability causes memory impairment. **Suggest one alternative hypothesis that could explain the correlation between missing teeth and hippocampus-dependent cognitive function.**

2B. [5 points, C]

2B. (i) Suppose that an experiment similar to the one described above is conducted, but instead of learning and retrieving words, subjects are asked to learn to draw a monkey while only seeing their hand and pen in a mirror. **Recalling a similar experiment from lecture, why would researchers conduct such an experiment?**

2B. (ii) Suppose that the number of attempts needed to correctly perform the task is recorded, while the subjects are paying full attention to their task (Table 2, next page). As in the word recall experiment, early and late conditions refer to the timing of gum chewing and the memory task. **Do these results suggest that mirror tracing is affected by chewing gum?** Justify your answers using your knowledge of statistics. Provide your answer on the next page.

Table 2. Results from a mirror tracing experiment, with subjects who chewed gum before the memory experiments (early or late), and subjects who did not chew gum prior to the experiments.

Experimental groups		Number of trials before correct drawing		
		Mean	SD	N
Early	Chewing gum	21	12	20
	Control	11	7	20
Late	Chewing gum	18	9	7
	Control	13	8	7

2B. (iii) According to Table 2, only seven subjects were included in the “Late” experiments. What would happen to the 95% confidence intervals of these measurements if more subjects were included? Explain in 1-2 sentences.

2C. [5 points, C]

In another study investigating the effect of gum chewing on attention, scientists obtained fMRI scans of subjects undergoing various attention tasks, while chewing gum or not. The researchers found that several brain regions were more active during the gum chewing condition than the non-chewing condition. Some of their results are shown in Figure 2B on the next page.

2C. (i) Describe the proxy used in fMRI in 1-2 sentences.

2C. (ii) Based on the information provided in Figure 2B below, estimate how many neurons are present in the active regions of the superior frontal gyrus (sfg). Note that there are three separate highlighted regions of the sfg. Assume that the average volume of a neuron cell body is $10^3 \mu\text{m}^3$, that neuron cell bodies comprise 3% of the volume of the cortex, and that the thickness of the cortex is 3 mm.

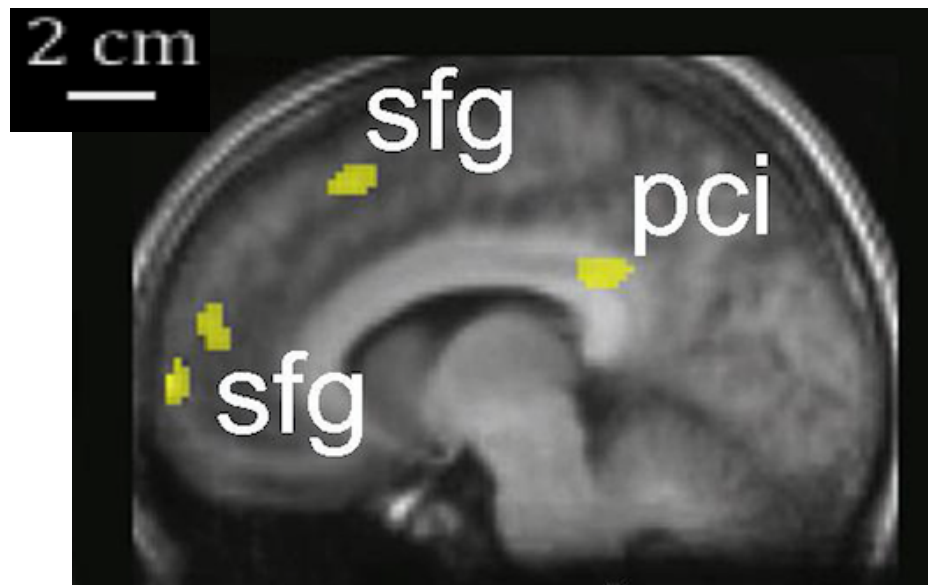


Figure 2B. Results from an fMRI experiment with participants chewing gum or not. The active regions are highlighted in yellow (dark gray in black and white versions).

2C. (iii) Were all the neurons in the highlighted parts of the sfg firing when the scan was taken? Explain in 1-2 sentences.

PHYSICS AND REALITY

Question 3. [15 points]

Trappist-1, a star with seven planets in orbit, is located 40 light-years from the Sun. Three of these planets have temperatures capable of supporting life as we know it. To search for life on the Trappist-1 planets, future scientists on Earth send an astronaut named Kunio to explore the Trappist-1 system; his spacecraft travels at $0.99c$ without changing direction.

3A. [5 points, C]

3A. (i) Calculate the relativistic gamma factor for Kunio's motion with respect to Earth.

3A. (ii) How many tubes of toothpaste should Kunio bring for his journey to Trappist-1? Consider only the journey *to* Trappist-1, not any time spent there or the return journey. Show all steps of your calculation, and clearly state all assumptions.

3B. [5 points, NC]

3B. (i) Traveling through the Solar System, Kunio sees the planet Jupiter directly out of his window. Which of the following pictures shown in Figure 3A below could represent what Kunio sees? Circle the correct answer.

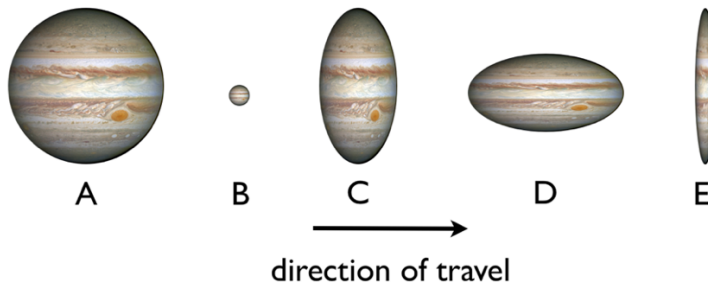


Figure 3A. Possible images of what Kunio sees, as he passes Jupiter at a speed of $0.99c$.

3B. (ii) During the journey, Kunio's friend Sonya remains on the Earth and eagerly awaits Kunio's return. Recall that Kunio is traveling at a constant velocity away from Earth. **Taking only the effects of special relativity into account, which of the following is true?** Circle the correct answer on the next page.

- a. According to Kunio, he (Kunio) is aging more quickly and according to Sonya, she (Sonya) is aging more quickly.
- b. According to Kunio, Sonya is aging more quickly, and according to Sonya, Kunio is aging more quickly.
- c. Kunio and Sonya agree that Kunio is aging more quickly.
- d. Kunio and Sonya agree that Sonya is aging more quickly.

3B. (iii) As mentioned above, during the journey, Kunio's friend Sonya remains on the Earth and eagerly awaits Kunio's return. Suppose that Kunio has left the Solar System, and is currently far away from any massive objects. **Taking only the effects of general relativity into account, which of the following is true?** Circle the correct answer below.

- a. According to Kunio, he (Kunio) is aging more quickly and according to Sonya, she (Sonya) is aging more quickly.
- b. According to Kunio, Sonya is aging more quickly, and according to Sonya, Kunio is aging more quickly.
- c. Kunio and Sonya agree that Kunio is aging more quickly.
- d. Kunio and Sonya agree that Sonya is aging more quickly.

3C. [5 points, C]

Energy is produced in the core of Trappist-1 during the process of fusion in which hydrogen atoms combine to form helium. In one second, 608.6 billion kilograms of hydrogen is converted into helium and Trappist-1 outputs 3.828×10^{26} joules of energy (units of J).

3C. (i) Given the above values, what percentage of the hydrogen mass is converted into energy inside Trappist-1?

3C. (ii) Compare this number to the efficiency of diesel engines, where just approximately 5×10^{-8} of the mass of the fuel is converted into energy. **By how many orders of magnitude is the efficiency of Trappist-1 different from that of diesel engines?**

3C. (iii) How does you answer in part (ii) above compare to the order of magnitude difference between the radius of a hydrogen atom and the radius of a proton?

Question 4. [15 points]

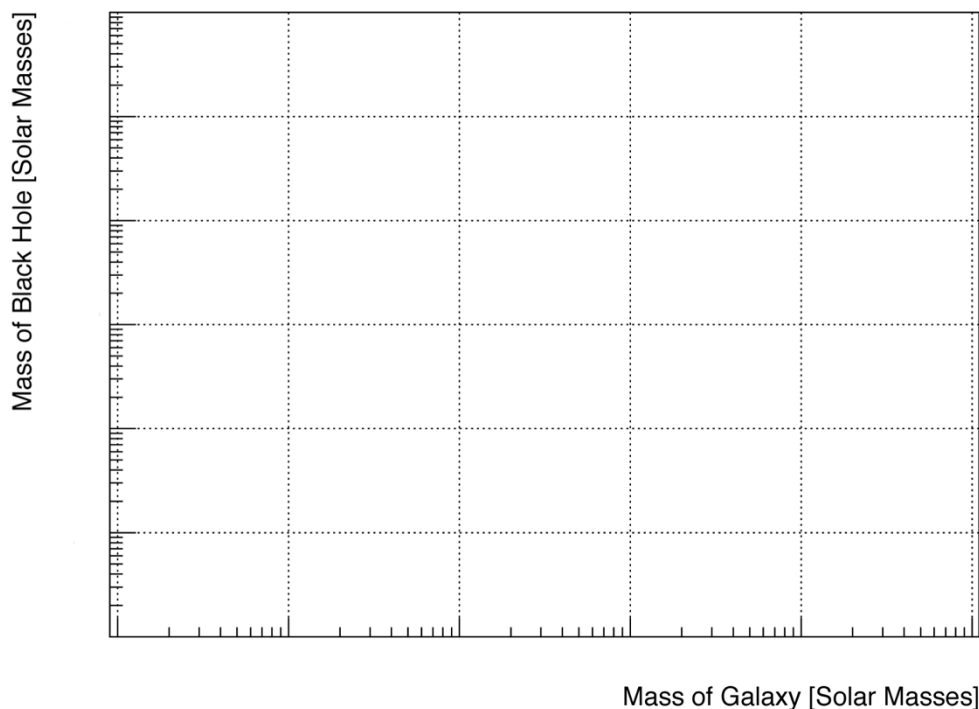
In recent years we have discovered that most galaxies contain a supermassive black hole at their center. The masses of four galaxies, and the masses of their supermassive black holes, are shown in Table 4. These masses are given in units of the mass of the sun.

4A. [5 points, SC]

4A. Plot the data for each of the four galaxies given in Table 4 on the log-log graph below. Plot the galaxy masses on the x-axis and the black hole masses on the y-axis. Be sure to label the major ticks of each axis first. Be sure to label each data point with the number (#) representing the galaxy.

***Table 4.** Galaxies, their masses and the masses of the black holes at their centers. Data adapted from Kormendy et al. (2013). One solar mass = 2×10^{30} kg.*

#	Galaxy Name	Galaxy Mass (in solar masses)	Black Hole Mass (in solar masses)
1	M32	1×10^9	2.5×10^6
2	NGC 3377	1×10^{10}	2×10^8
3	NGC 1399	1×10^{11}	8×10^8
4	M87	1×10^{12}	6×10^9



4B. [5 points, C]

4B. (i) Using the data in Table 4, what is the Schwarzschild radius of the supermassive black hole in M87?

4B. (ii) Assume a black hole is a sphere with a radius equal to its Schwarzschild radius. What is the density of the supermassive black hole in M87?

4B. (iii) How does the density of the M87 black hole compare with the density of air? (*The density of air is $\sim 1 \text{ kg/m}^3$*)

4C. [5 points, C]

Scientists at the CERN laboratory in Geneva, Switzerland discovered the Higgs particle in 2012, and determined that its mass is equal to $2.24 \times 10^{-25} \text{ kg}$.

4C. (i) If you can determine the position of the Higgs particle in the particle accelerator to within 0.1 nm, what is the minimum uncertainty with which you can determine its velocity? Give your answer in units of miles/hour.

4C. (ii) What can you say about the location of the Higgs particle before it is measured?